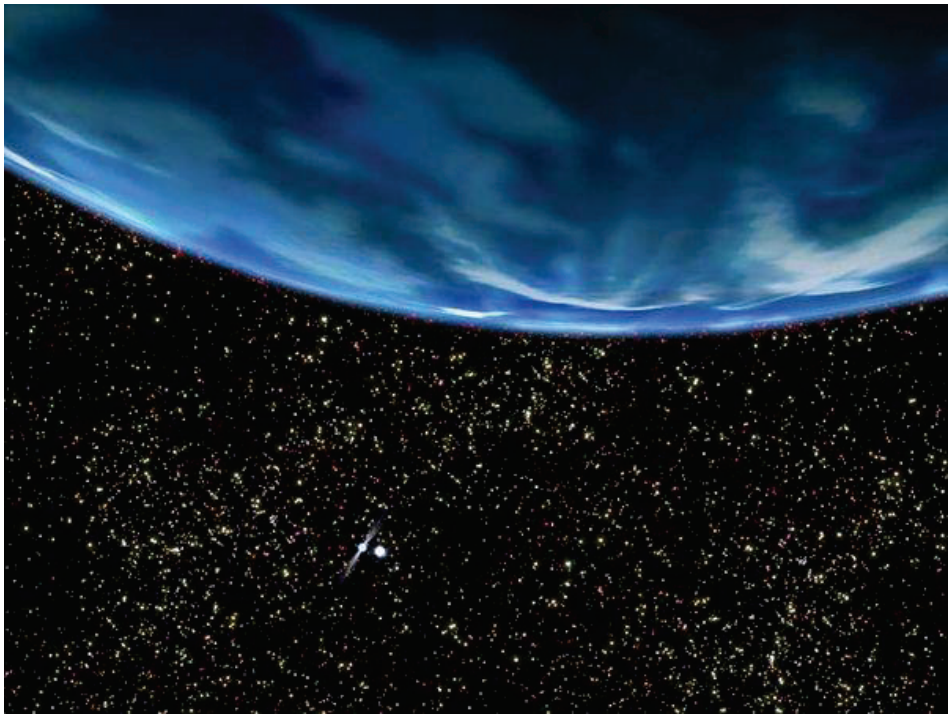
Instruction: Use the text in accomplishing Activity 1- *You Can Fill It!*

Started with a Bang

Origin and Structure of the Earth



Photograph courtesy NASA/Brad Hansen (UCLA)/Harvey Richer (UBC)/Steinn Sigurdsson (Penn State)/Ingrid Stairs (UBC)/Stephen Thorsett (UCSC)

The most popular theory of our Universe's origin centers on a cosmic cataclysm unmatched in all of history—the **Big Bang**. This theory was born of the observation that other galaxies are moving away from our own at great speed, in all directions, as if they had all been propelled by an ancient explosive force.

Before the big bang, scientists believe, the entire vastness of the observable universe, including all of its matter and radiation, was compressed into a hot, dense mass just a few millimeters across. This nearly incomprehensible state is theorized to have existed for just a fraction of the first second of time.

Big bang proponents suggest that some 10 billion to 20 billion years ago, a massive blast allowed all the universe's known matter and energy—even space and time themselves—to spring from some ancient and unknown type of energy.

The theory maintains that, in the instant—a trillion-trillionth of a second—after the big bang, the universe expanded with incomprehensible speed from its pebble-size origin to astronomical scope. Expansion has apparently continued, but much more slowly, over the ensuing billions of years.

Scientists can't be sure exactly how the universe evolved after the big bang. Many believe that as time passed and matter cooled, more diverse kinds of atoms began to form, and they eventually condensed into the stars and galaxies of our present universe.

Origins of the Theory



Georges Lemaître
Retrieved from www.findagrave.com



Edwin Hubble
Retrieved from www.stsci.edu

A Belgian priest named Georges Lemaitre first suggested the Big Bang Theory in the 1920s when he theorized that the universe began from a single primordial atom. The idea subsequently received major boosts by Edwin Hubble's observations that galaxies are speeding away from us in all directions, and from the discovery of **cosmic microwave radiation** by Arno Penzias and Robert Wilson.

The glow of cosmic microwave background radiation, which is found throughout the universe, is thought to be a tangible remnant of leftover light from the big bang. The radiation is similar to that used to transmit TV signals via antennas, but it is the oldest radiation known and may hold many secrets about the universe's earliest moments.

The Big Bang Theory leaves several major questions unanswered. One is the original cause of the Big Bang itself. Several answers have been proposed to address this fundamental question, but none has been proven—and even adequately testing them has proven to be a formidable challenge.

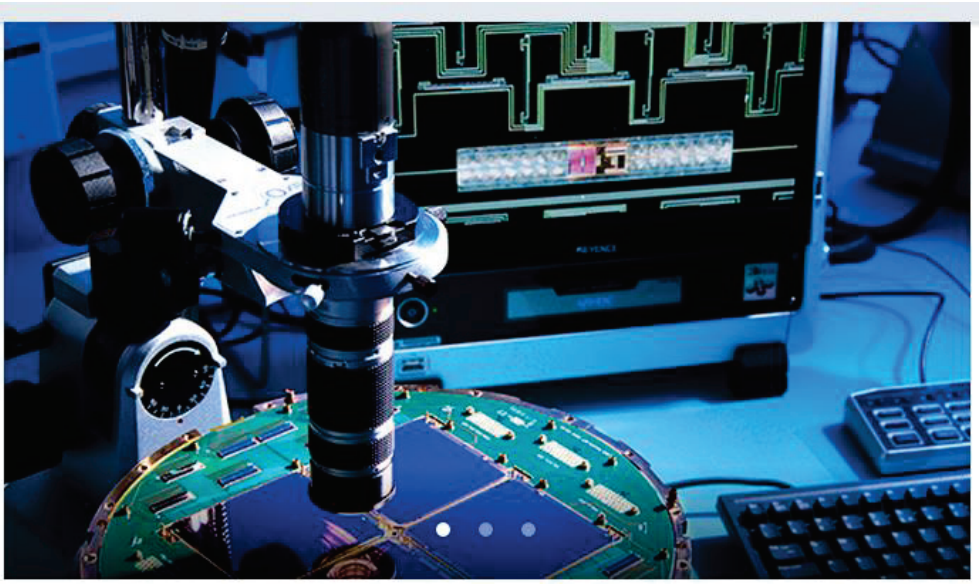
Reference:

Origin of the Universe. (n.d.). Retrieved from science.nationalgeographic.com: http://science.nationalgeographic.com/science/space/universe/origins-universe-article/

Instruction: Use the text in accomplishing Activity 1- *You Can Fill It!*

NASA Technology Views Birth of the Universe

Origin and Structure of the Earth



The BICEP2 telescope at the South Pole used a specialized array of superconducting detectors to capture polarized light from billions of years ago. Image credit: NASA/JPL-Caltech

Astronomers are announcing today that they have acquired the first direct evidence that gravitational waves rippled through our infant universe during an explosive period of growth called inflation. This is the strongest confirmation yet of cosmic inflation theories, which say the universe expanded by 100 trillion times, in less than the blink of an eye.

The findings were made with the help of NASA-developed detector technology on the BICEP2 telescope at the South Pole, in collaboration with the National Science Foundation.

"Operating the latest detectors in ground-based and balloon-borne experiments allows us to mature these technologies for space missions and, in the process, make discoveries about the universe," said Paul Hertz, NASA's Astrophysics Division director in Washington.

Our universe burst into existence in an event known as the Big Bang 13.8 billion years ago. Moments later, space itself ripped apart, expanding exponentially in an episode known as inflation. Tell-tale signs of this early chapter in our universe's history are imprinted in the skies, in a relic glow called the cosmic microwave background. Recently, this basic theory of the universe was again confirmed by the Planck satellite, a European Space Agency mission for which NASA provided detector and cooler technology.

But researchers had long sought more direct evidence for inflation in the form of gravitational waves, which squeeze and stretch space.

"Small, quantum fluctuations were amplified to enormous sizes by the inflationary expansion of the universe. We know this produces another type of waves called density waves, but we wanted to

test if gravitational waves are also produced," said project co-leader Jamie Bock of NASA's Jet Propulsion Laboratory, Pasadena, Calif., which developed the BICEP2 detector technology. Bock has a joint appointment with the California Institute of Technology, also in Pasadena.

The gravitational waves produced a characteristic swirly pattern in polarized light, called "B-mode" polarization. Light can become polarized by scattering off surfaces, such as a car or pond. Polarized sunglasses reject polarized light to reduce glare. In the case of the cosmic microwave background, light scattered off particles called electrons to become slightly polarized.

The BICEP2 team took on the challenge to detect B-mode polarization by pulling together top experts in the field, developing revolutionary technology and traveling to the best observing site on Earth at the South Pole. The collaboration includes major contributions from Caltech; JPL; Stanford University, Stanford, Calif.; Harvard University, Cambridge, Mass.; and the University of Minnesota, Minneapolis.

As a result of experiments conducted since 2006, the team has been able to produce compelling evidence for the B-mode signal, and with it, the strongest support yet for cosmic inflation. The key to their success was the use of novel superconducting detectors. Superconductors are materials that, when chilled, allow electrical current to flow freely, with zero resistance.

"Our technology combines the properties of superconductivity with tiny structures that can only be seen with a microscope. These devices are manufactured using the same micro-machining process as the sensors in cell phones and Wii controllers," said Anthony Turner, who makes these devices using specialized fabrication equipment at JPL's Micro devices Laboratory.

The B-mode signal is extremely faint. In order to gain the necessary sensitivity to detect the polarization signal, Bock and Turner developed a unique array of multiple detectors, akin to the pixels in modern digital cameras but with the added ability to detect polarization. The whole detector system operates at a frosty 0.25 Kelvin, just 0.45 degrees Fahrenheit above the lowest temperature achievable, absolute zero.

"This extremely challenging measurement required an entirely new architecture," said Bock. "Our approach is like taking a camera and building it on a printed circuit board."

The BICEP2 experiment used 512 detectors, which sped up observations of the cosmic microwave background by 10 times over the team's previous measurements. Their new experiment, already making observations, uses 2,560 detectors.

These and future experiments not only help confirm that the universe inflated dramatically, but are providing theorists with the first clues about the exotic forces that drove space and time apart.

The results of this study have been submitted to the journal Nature.

JPL is managed by the California Institute of Technology in Pasadena for NASA.

Reference:

(2014, March 17). Retrieved from www.jpl.nasa.gov: http://www.jpl.nasa.gov/news/news.php?release=2014-082

Instruction: Use the text in accomplishing Activity 1- *You Can Fill It!*

How Did the Sun and Planetary System Form?

Origin and Structure of the Earth

Theories about the origin of the Solar System

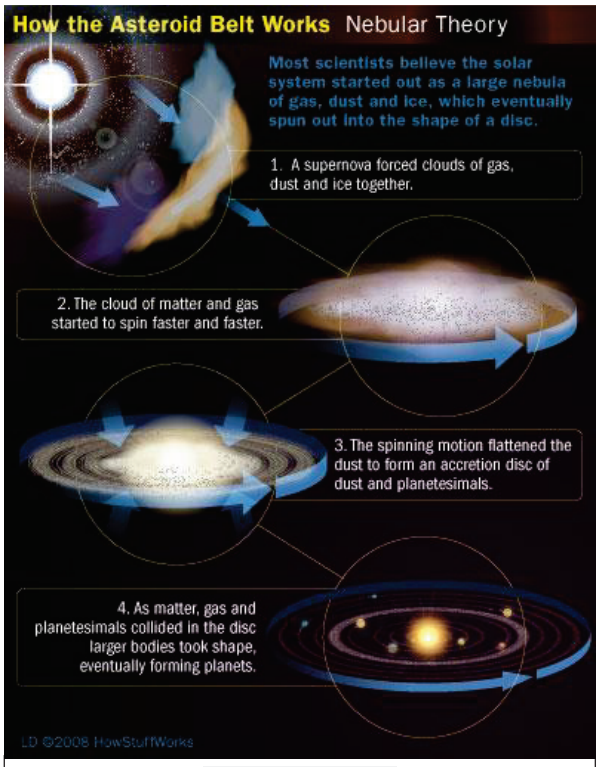


Nicolaus Copernicus in the 1000 Zloty bill of Republic of Poland
Retrieved from www.tcocd.de

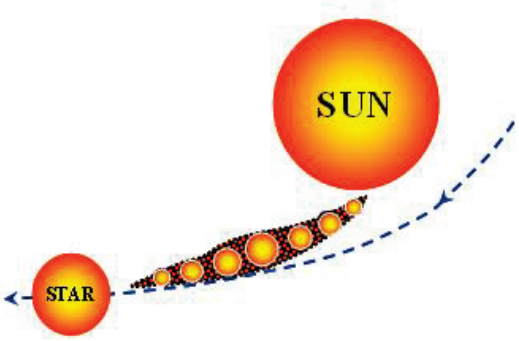
The Sun and the celestial bodies that revolve around it, including the planets with their satellites, comets, and meteors, comprise the Solar System. Before the modern heliocentric or sun-centered system of planetary motions was described by Nicolaus Copernicus in 1543, the earth has been assigned the central place in the universe. The most enduring arrangement of this geocentric earth-centered plan was the Ptolemaic System. Copernicus, however, placed the planets in circular paths around the sun and placed the earth as the sixth planet with its satellite, the moon.

On the other hand, there were some scientists who continued studying the origin of the solar system. They have formulated theories to prove and explain the origin of the solar system.

In 1755, a German philosopher, Immanuel Kant, in 1755, and a French mathematician, Laplace, in 1796, published closely related hypotheses which postulated that the solar system was derived from the condensation of an enormously dispersed gaseous atmosphere surrounding the sun. Increased rotational velocity of this atmosphere during condensation was assumed to have produced a discoidal shape, the plane of the disk coinciding with that of the sun's equator. When velocity reached a critical point, centrifugal force would throw off part of the gas as a ring and materials of each ring gradually assembled into a gaseous globe, which eventually became a solid planet revolving around the sun in a circular orbit like that of the ring from which it was formed. While these gaseous globes were contracting, most of them abandoned the rings, which assembled into satellites revolving in circles around these planets. Thus according to the hypothesis called the Nebular Hypothesis, the solar system developed with the observed regularities in its motions. When other discrepancies seemed to invalidate the hypothesis of rotational instability, a second theory, the Planetesimal Theory, was formulated. It supposed that the planetary system was formed from materials removed from the sun by tidal action caused by a passing star. The projection became masses of gases which revolved around the sun. The larger bodies attracted smaller ones and became planets. The smaller bodies became asteroids, meteors, and satellites of planets.



Nebula Theory
Retrieved from science.howstuffworks.com



Planetesimal Theory
Retrieved from www.daviddarling.info

Another theory was formulated by the German physicist Carl Friedrich von Weizacken and U.S. chemist Harold C. Urey, called the Dust Cloud Theory, in 1945. The nebula was assumed to have a composition mainly of hydrogen and helium, like the sun, with only 1% of heavier elements. The mass of this "Dust Cloud" was originally 10 % of the sun's mass or about a hundred times as great as the present combined mass of the planets and satellites. The nebula was much flattened by its rotation, which as of the planetary type in that the gas molecules moved faster as they were closer to the sun. Interactions of the gas molecules accelerated the lighter ones so that they mostly escaped from the nebula. The interactions also produced swirls, forming lumps in the nebula that could grow to become planets and satellites.

Another hypothesis, the Protoplanet Hypothesis, was proposed in 1949 by Gerald P. Kuoper. It states that the original nebula was so massive that on further contraction and flattening, it broke into separate clouds or protoplanets. These remained stable in the tidal field of the sun. As they contracted, they developed denser cores surrounded by large atmospheres of the lighter gases. Later the shrinking primitive sun became hot enough to emit powerful corpuscular and ultraviolet radiations. The radiations drove away into space remnants of the nebula and the large atmosphere of the planets, which thus looked like a swarm of comets with tails. This hypothesis proposed a process that could have developed planetary systems around many stars. Indeed, it has been surmised that the majority of yellow star, like the sun, may possess systems of planets.

The solar system where the earth belongs is just a small part of the vastness that we call the universe. No one knows its boundaries nor its origin. Did it originate suddenly in an enormous explosion or has it been in a process of creation without a definite beginning or end? What were mentioned above are just theories which need to be proven.

Reference:

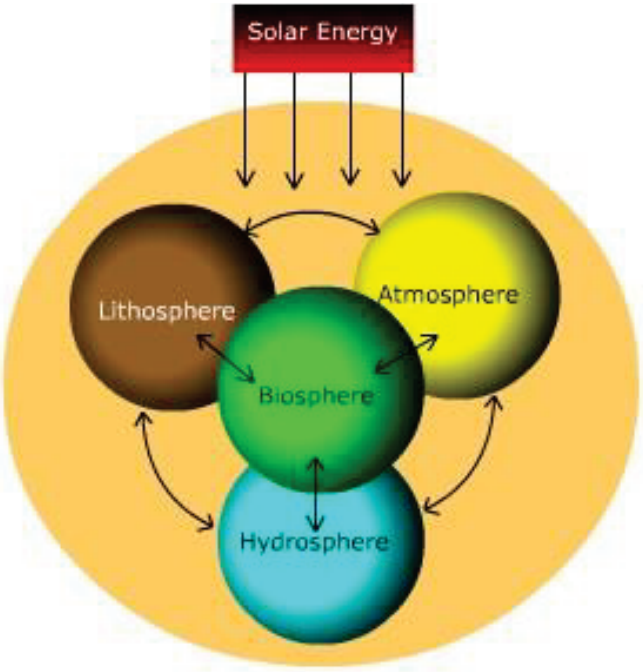
Braganza, M. C. (1997). *Earth Science.* Quezon City: Rex Printing Company.

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Subsystems of the Earth

Origin and Structure of the Earth

The Earth is comprised of four parts namely Lithosphere, Hydrosphere, Biosphere, and Atmosphere. Their names are derived from the Greek words for stone (litho), water (hydro), air (atmo), and life (bio).



Subsystems of the Earth
cladenver.esfaculty.ucdu

Lithosphere

The lithosphere is basically the rocky crust of the earth. It is inorganic and is composed mainly of different kinds of minerals.

This is the outer part of the Earth, including both the uppermost mantle and the crust. The mantle surrounds the core and lies beneath the crust. The physical characteristics of the mantle vary with the depth. Near the surface, to a depth of about 100 kilometers, the outermost mantle is cool, strong, and hard. In contrast, the layer below 100 kilometers is so hot that the rock is weak, soft, plastic, and flows slowly-like cold honey. Even deeper, pressure overwhelms temperature and the mantle rock becomes strong again. The crust is the outermost, a thin veneer below a layer of soil and beneath the ocean water. It is composed almost entirely of solid rock.

Hydrosphere

This is composed of all the waters, which circulates in the Earth. This includes the oceans, seas, rivers, lakes, and even the moisture in the air. Oceans cover 71 percent of Earth and its currents transport heat across vast distances, altering global climate.

Atmosphere

Atmosphere is the mass of air surrounding our planet. It is subdivided into different layers of different densities. The air of Earth is comprised of 79% Nitrogen and fewer than 21% Oxygen. The remaining amount is shared by Carbon Dioxide and other form of gasses. Atmosphere is held to Earth by gravity

and thins rapidly with altitude. Ninety-nine percent is concentrated in the first 30 kilometers, but a few traces remain as far as 10,000 kilometers above the Earth's surface.

The atmosphere supports life because animals need oxygen, and plants need both carbon dioxide and oxygen. In addition, the atmosphere supports life indirectly by regulating climate. Air acts as both blanket and a filter, retaining heat at night and shielding us from direct solar radiation during the day. Wind transports heat from the equator toward the poles, cooling equatorial regions, and warming temperature and polar zones.

Biosphere

The biosphere comprises of all living organisms, from the smallest bacteria to the largest whale. Plants, animals, and single-celled organism are all part of the biosphere. Biosphere includes the uppermost geosphere, the hydrosphere, and the lower parts of the atmosphere. Sea life concentrates near the surface, where sunlight is available. Plants also grow on Earth's surface, with roots penetrating a few meters underground. Large populations of bacteria live in rock to depths of as much as 4 kilometers, some organisms live on the ocean floor, and a few windblown microorganisms drift at heights of 10 kilometers or more. But even at these extremes, the biosphere is a very thin layer at Earth's surface. Plants and animals are clearly affected by Earth's environment. Organisms breathe air, require water, and thrive in a relatively narrow temperature range. Terrestrial organisms ultimately depend on soil, which is part of the geosphere. Less obviously, plants and animals also alter and form the environment they live in. For example, living organisms contributed to the evolution of the modern atmosphere.

These subsystems interact with each other and influences the climate, trigger geological processes, and affect life all over the Earth.

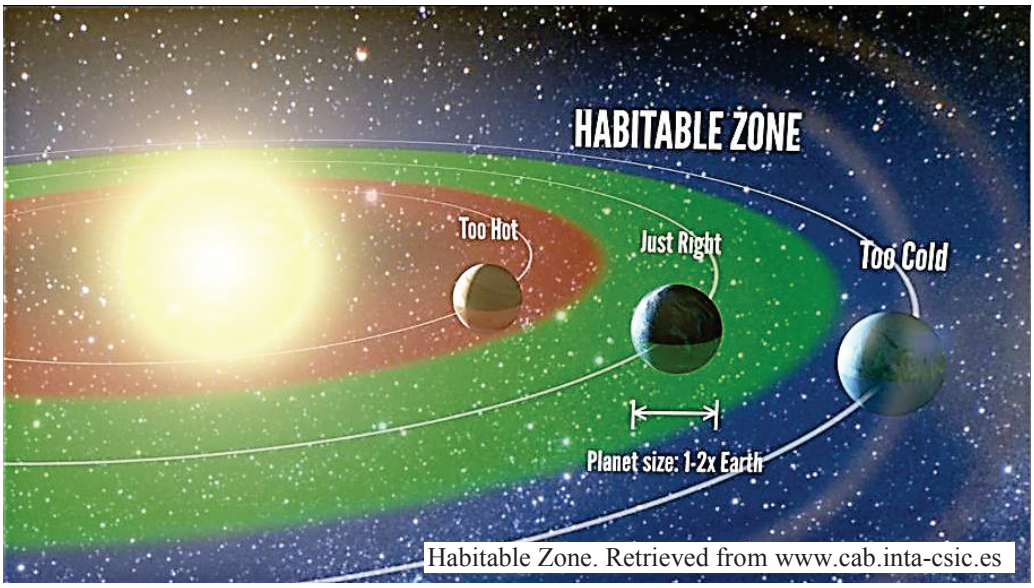
Reference:

Earth and Life Science Instructor's Guide. (2015). *Origin and Structure of the Earth and Beyond* . Rizal, Philippines: STI Academic Center.

Instruction: Use the text in accomplishing Activity 1- *You Can Fill It!*

The Characteristics of Earth that are Necessary to Support Life

Origin and Structure of the Earth



Habitable Zone. Retrieved from www.cab.inta-csic.es

There are a few key ingredients that scientists often agree are needed for life to exist — but much debate remains as to what limits there actually might be on life. Even Earth hosts some strange creatures that live in extreme environments

Here's what makes life able to thrive on our home planet (and likely for alien life to arise on other worlds):

Water

"First, you'd need some kind of liquid, any place where molecules can go react," Seager told OurAmazingPlanet. In such a soup, the ingredients for life as we know it, such as DNA and proteins, can swim around and interact with each other to carry out the reactions needed for life to happen.

The most common contender brought up for this solvent is the one life uses on Earth: water. Water is an excellent solvent, capable of dissolving many substances. It also floats when it is frozen, unlike many liquids, meaning that ice can insulate the underlying fluid from freezing further. If water instead sunk when frozen, this would allow another layer of water to freeze and sink, and eventually all the water would get frozen, making the chemical reactions behind life impossible.

Astronomers looking for extra-terrestrial life most often focus on planets in the so-called habitable zones of their stars — orbits that are neither too hot nor too cold for liquid water to persist on the surfaces of those worlds. Earth happened to hit the Goldilocks mark, forming within the sun's habitable zone. Mars and Venus lie outside it; if Earth's orbit had been just a bit further inside or outside of where it is, life may likely never have arisen and the planet would be a cold desert like Mars or a cloudy furnace like Venus.

Of course, alien life may not play by the rules we're used to on Earth.

Astrobiologists increasingly suggest looking beyond conventional habitable zones. For instance, while liquid water might not currently persist on the surface of Mars or Venus, there may have been a time when it did. Life might have evolved on their surfaces in that time, and then either fled to safer locales on those planets, such as underground, or adapted to the environment when it became harsh, much as so-called extremophile organisms have on Earth, or both.

In addition, other solvents might host life. "Saturn's moon, Titan, has liquid methane and ethane." Seager said.

Energy

Second, life needs energy. Without energy, virtually nothing would happen.

The most obvious source of energy is a planet or moon's host star, as is the case on Earth, where sunlight drives photosynthesis in plants. The nutrients created by photosynthesis in turn are what the bulk of life on Earth directly or indirectly relies on for fuel.

Still, countless organisms on Earth subsist on other sources of energy as well, such as the chemicals from deep water vents. There may be no shortage of energy sources for life to live off.

Time

Scientists have argued that habitable worlds need stars that can live at least several billion years, long enough for life to evolve, as was the case on Earth.

Some stars only live a few million years before dying. Still, "life might originate very fast, so age is not that important," astrobiologist Jim Kasting at Pennsylvania State University told OurAmazingPlanet.

For instance, the Earth is about 4.6 billion years old. The oldest known organism first appeared on Earth about 3.5 billion years ago, meaning that life might conceivably evolve in 1.1 billion years or less. However, more complex forms of life did take longer to evolve — the first multicellular animals did not appear on Earth until about 600 million years ago. Because our sun is so long-lived, comparatively, higher orders of life, including humans, had time to evolve.

Recycling

Other researchers have suggested that plate tectonics is vital for a world to host life — that is, a planet whose shell is broken up into plates that constantly move around.

"People talk about plate tectonics as essential in recycling molecules life needs," Seager said. For instance, carbon dioxide helps trap heat from the sun to keep Earth warm. This gas normally gets bound up in rocks over time, meaning the planet would eventually freeze. Plate tectonics helps ensure this rock gets dragged downward, where it melts, and this molten rock eventually releases this carbon dioxide gas back into the atmosphere through volcanoes.

"Plate tectonics is useful but probably not imperative," Kasting said. Seager agreed, saying that "volcanism might very well provide enough fresh supplies of whatever life might need."

Bonus features

Other factors researchers have trotted out for why life succeeded on Earth include how little variation there is in our sun's radiation compared with more volatile stars, or how our planet has

a magnetic field that protects us from any storms of charged particles from the sun. Violent bursts of radiation could have scoured life from Earth in its early, fragile stages.

Still, "people are constantly rethinking each of these things and how important they are," Seager said. "We're trying to be less conservative and more open-minded. We want to learn about what gray areas might exist for possible life."

Earth remains the only known planet to host life, due to a unique combination of factors. However, continued monitoring of alien worlds might one day change that, by finding other planets that share these attributes or by discovering other ways that life has found to blossom in the universe.

Habitable Planets

Even if planets exist near some other stars, there is no guarantee that they are habitable. Astronomers have proposed several conditions needed to make a planet habitable:

- The central star should not be more than 1.5 Mo (Solar mass) so that it will last long enough for substantiated life to evolve at least 2 billion years and will not kill evolving life with too much UV radiation which breaks down organic molecules.
- The central star should be at least 0.3 Mo to be warm enough to create a large orbital zone in which a planet could retain liquid water.
- The central star should not flare violently or emit strong x-rays. It should be on the main sequence in order to be stable, long enough to give its planet long-term climatic stability.
- The planet must orbit at the right distance from the star so that liquid water will neither evaporate nor permanently freeze.
- The planet’s orbit must be circular and stable enough to keep it at a proper distance and prevent drastic seasonal changes.
- The planet’s gravity must be strong enough to hold a substantial atmosphere.

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